NLP Language Modelling exercises

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Q1 Consider the following corpus of 3 sentences:

- <s> I am Sam </s>
- <s> Sam I am </s>
- <s> I saw a unicorn </s>

Q1.1 Compute all bigram probabilies by filling table 1 where the rows correspond to the first word of the bigram (w_{n-1}) and the columns correspond to the second word (w_n).

	I	am	Sam	saw	a	unicorn	</s>
<s>							
I							
am							
Sam							
saw							
a							
unicorn							

Table 1: Bigram probabilities

Q1.2 Based on these probabilities, compute the probability of the following sentences:

- <s> I am </s>
- <s> I am a unicorn </s>
- <s> I saw a whale </s>

Q1.3 Generate a sentence using our bigram model, selecting the most probable bigram at each step (you choose how to break ties).

Q1.4 ✏️ Smoothe your bigram probabilities with add-one smoothing from the earlier exercise. Recompute the probabilities for the sentences of **Q1.2**.

🤔 How are the probabilities affected by smoothing? Would add-k with a value of k between 0 and 1 be better? Why? (It can be informative to try this exercise with other values of k to form a clearer intuition).

Q1.5 ✏️ Calculate the interpolated probabilites of the sentences of **Q1.2** with the bigram model of **Q1.1**, a MLE unigram model, and $\lambda_1 = \lambda_2 = 0.5$.

🤔 How do the results compare to add-one smoothing? How would altering the λ weights alter the results? (It can be informative to try this exercise with other λ weights).